Zander Barajas

858-525-2267 | zanderbr@protonmail.com | linkedin.com/in/xzanderbr | github.com/xZanderBR | xzanderbr.github.io

Education

San Diego State University Aug. 2024 – May 2026 (Expected)
Bachelor of Science in Computer Science *SDSU GPA: 3.8*

San Diego Mesa College Sept. 2021 – May 2024
Associate's Degree for Transfer in Computer Science

Experience

Software Engineering Intern Summers 2024 & 2025
Envision Engineering Inc *San Diego, CA*

- **Summer 2025:** Led the design and deployment of a full-stack web app (Next.js, FastAPI, PostgreSQL, Docker) with GitHub Actions CI/CD on Vercel/Render, including a React/Tailwind dashboard with Chart.js to visualize project deliverables, cutting manual reporting time from around 1-2 hours per week to under 15 minutes
- **Summer 2024:** Coordinated with engineers to automate CAD workflows with Python (pyautocad, Revit API), reducing drawing cleanup, layer management, and export time by around 70% across 25+ projects; also deployed and maintained a NAS server to improve project sharing and accessibility

Projects

Powerlifting Meet Calculator | *C/C++, CMake, ImGui, Vulkan, Google Test, Git* Sept. 2024 – Nov. 2024

- Developed a cross-platform calculator with data sorting, JSON import/export, lifter management, and score calculations
- Built a smooth and responsive graphical user interface utilizing ImGui with Vulkan for rendering
- Implemented asynchronous task handling and a custom multithreaded sorting method, improving performance by over 750% for large datasets
- Automated builds and testing on multiple platforms using CMake and GitHub Actions

Aztec List | *Python, FastAPI, React, Next.js, TypeScript, PostgreSQL, Docker, Git* September 2025 – Present

- Collaborated in a team of five to build a full-stack marketplace platform for college students with listing management, messaging, and admin moderation
- Developed RESTful API with FastAPI and SQLAlchemy ORM, and built Next.js frontend with React, TypeScript, and Tailwind CSS
- Implemented JWT authentication, Argon2 password hashing, rate limiting, and image upload supporting up to 10 images per listing
- Containerized with Docker Compose and established comprehensive testing with pytest, Jest, and Playwright

Drone 2D Obstacle Avoidance | *Python, Gymnasium, PyGame, Stable-Baselines3, PPO* Mar. 2025 – Present

- Led a group of four to design and implement a custom Gymnasium environment in PyGame for drone pathfinding and obstacle avoidance, setting up the GitHub repository and project structure
- Spearheaded major components including training pipeline design, evaluation scripts, and TensorBoard logging, ensuring consistent workflows across the team
- Trained PPO agent using Stable-Baselines3 with VecNormalize, Monitor, and custom reward shaping, achieving up to 72% goal success rate after 500k steps

Technical Skills

Concepts: Data Structures & Algorithms, OOP, Concurrency, Operating Systems, AI/ML, Full-Stack Development, CI/CD Pipelines, REST APIs, Cloud Deployment
Languages: C/C++, C#, Python, Java, Rust, JavaScript, TypeScript, HTML/CSS
Frameworks/Libraries: .NET, Next.js, React, FastAPI, Tailwind CSS, Vulkan, Stable-Baselines3, pytest, Jest
Databases: PostgreSQL, MySQL/MariaDB, MongoDB
Tools: Git, Docker, GitHub Actions, Linux, AWS, Azure, Google Cloud, Vercel/Render